Akoizo Original Practice - Form 1 - Answer key

Division B | Remote Sensing | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) What does a sensor directly record in optical imaging?

Unit: Remote sensing foundations and history.
Worked-case data: An image becomes brighter after a display stretch, but its stored reflectance values are unchanged. Has surface reflectance increased?

A. A band-dependent energy response

B. Tree species with certainty

C. Human land-use intent

D. A complete climate mechanism

Correct: A

2. (2 points) What is the best next interpretation step?

Unit: Passive images and active radar.
Worked-case data: Weather: Clouds cover a flooded valley. Optical image: The ground cannot be seen beneath the clouds. Radar image: New dark areas follow the low valley floor.
Separate investigation observations:
Weather: Clouds cover a flooded valley.
Optical image: The ground cannot be seen beneath the clouds.
Radar image: New dark areas follow the low valley floor.

A. Compare radar patterns with elevation and independent flood observations

B. Assume thermal imagery always sees through clouds

C. Discard radar because it requires sunlight

D. Label every dark pixel as water without checking

Correct: A

3. (2 points) Which observation strategy best follows rapid storm development?

Unit: Choosing an orbit for an observation.
Worked-case data: Task: Track a storm every few minutes. Sensor A: Views one broad region continuously. Sensor B: Collects a detailed strip on occasional passes.
Separate investigation observations:
Task: Track a storm every few minutes.
Sensor A: Views one broad region continuously.
Sensor B: Collects a detailed strip on occasional passes.

A. Ignore the time between images

B. Assume all polar satellites continuously view the same place

C. Use only one detailed image from sensor B

D. Use the frequent regional views from sensor A

Correct: D

4. (2 points) What determines false-color appearance?

Unit: Composites, spectral signatures, and contrast.
Worked-case data: NIR is assigned to the red display channel. A field has NIR = 0.6 and red reflectance = 0.1. Explain its likely display and limitation.

A. Only sensor altitude

B. Only file compression

C. Band-to-channel assignment

D. Only the object name

Correct: C

5. (2 points) Area of a square 30 m pixel?

A. 900 m^2

B. 30 m^2

C. 60 m^2

D. 300 m^2

Correct: A

6. (2 points) What is the average percent slope between these sites?

Unit: Map scale, elevation, and land-cover evidence.
Worked-case data: Scale: 1 cm represents 500 m. Map distance: Two sites are 4 cm apart. Elevation: They differ by 100 m.
Separate investigation observations:
Scale: 1 cm represents 500 m.
Map distance: Two sites are 4 cm apart.
Elevation: They differ by 100 m.

A. 5%

B. 50%

C. 500%

D. 25%

Correct: A

7. (2 points) What is the range to the surface?

Unit: Light, temperature, and radar distance.
Worked-case data: Pulse speed: 300,000,000 m/s. Round-trip time: 0.004 s. Geometry: A pulse travels straight to the surface and back.
Separate investigation observations:
Pulse speed: 300,000,000 m/s.
Round-trip time: 0.004 s.
Geometry: A pulse travels straight to the surface and back.

A. 300,000 m

B. 75,000,000,000 m

C. 1,200,000 m

D. 600,000 m

Correct: D

8. (2 points) Which statement is supported?

Unit: Albedo, surface temperature, and image comparisons.
Worked-case data: Pavement: Dark visible image; thermal scale indicates 42°C. Trees: Thermal scale indicates 28°C. Acquisition: Same cloud-free scene at the same time.
Separate investigation observations:
Pavement: Dark visible image; thermal scale indicates 42°C.
Trees: Thermal scale indicates 28°C.
Acquisition: Same cloud-free scene at the same time.

A. Albedo alone proves the entire temperature difference

B. Every dark surface is always warmer

C. The pavement surface is warmer in this observation

D. The air above the pavement must be 42°C

Correct: C

9. (2 points) Which interpretation has the strongest support?

Unit: Land-cover change, hazards, and classification.
Worked-case data: Before: A field is mapped as vegetation. After: The field is dark in a cloudy optical image. Radar and ground report: Both support standing water at that location.
Separate investigation observations:
Before: A field is mapped as vegetation.
After: The field is dark in a cloudy optical image.
Radar and ground report: Both support standing water at that location.

A. A dark optical pixel alone proves flooding

B. Flooding is supported by multiple independent observations

C. Clouds cannot affect optical observations

D. A classification map never needs validation

Correct: B

10. (2 points) What can explain lower field NDVI without a unique disease diagnosis?

A. A changed notebook font

B. A higher map title

C. Greater exposed-soil fraction

D. A renamed file only

Correct: C

11. (2 points) Why can a field plot disagree with a pixel?

Unit: Remote sensing foundations and history.
Worked-case data: An image becomes brighter after a display stretch, but its stored reflectance values are unchanged. Has surface reflectance increased?

A. Calibration removes all mixtures

B. They sample different spatial supports

C. Pixels have no ground area

D. Field measurements are always exact

Correct: B

12. (4 points) Explain the evidence supporting your answer, including the calculation or comparison.

Unit: Passive images and active radar.
Worked-case data: Weather: Clouds cover a flooded valley. Optical image: The ground cannot be seen beneath the clouds. Radar image: New dark areas follow the low valley floor.
Separate investigation observations:
Weather: Clouds cover a flooded valley.
Optical image: The ground cannot be seen beneath the clouds.
Radar image: New dark areas follow the low valley floor.

answer: The optical view is obstructed. Radar supplies different evidence, but dark returns are not unique to flooding. Use elevation and independent observations to test the flood interpretation. (4 points).

13. (2 points) Independent stretches can make date comparisons misleading because…

A. Equal colors may represent unequal values

B. All stored values are erased

C. NIR becomes visible light physically

D. Registration is no longer needed

Correct: A

14. (2 points) How many ideal 8-bit levels exist?

A. 64

B. 4096

C. 256

D. 8

Correct: C

15. (2 points) Validation data should be…

Unit: Applied image interpretation and agriculture.
Worked-case data: A field pixel changes from pure vegetation to 50% vegetation and 50% soil using NIR/red pairs (0.6,0.1) and (0.3,0.2).

A. Independent of threshold fitting

B. The same selected examples only

C. Chosen only where predictions are correct

D. Discarded after fitting

Correct: A

16. (2 points) Which metadata is essential for date comparison?

Unit: Remote sensing foundations and history.
Worked-case data: An image becomes brighter after a display stretch, but its stored reflectance values are unchanged. Has surface reflectance increased?

A. Only file name length

B. Only border color

C. Only display size

D. Product units and acquisition conditions

Correct: D

17. (4 points) Explain “Passive sensor” in your own words.

Unit: Passive images and active radar.
Worked-case data: Weather: Clouds cover a flooded valley. Optical image: The ground cannot be seen beneath the clouds. Radar image: New dark areas follow the low valley floor.
Separate investigation observations:
Weather: Clouds cover a flooded valley.
Optical image: The ground cannot be seen beneath the clouds.
Radar image: New dark areas follow the low valley floor.

answer: Detects energy supplied by another source. (4 points).

18. (4 points) Explain “Orbit” in your own words.

Unit: Choosing an orbit for an observation.
Worked-case data: Task: Track a storm every few minutes. Sensor A: Views one broad region continuously. Sensor B: Collects a detailed strip on occasional passes.
Separate investigation observations:
Task: Track a storm every few minutes.
Sensor A: Views one broad region continuously.
Sensor B: Collects a detailed strip on occasional passes.

answer: A path around a body under gravity. (4 points).